

issues by providing extremely fast access to a scalable network of flash based storage devices, and to provide a platform for application-specific hardware accelerators on the data path on each of the storage devices.

According to the project overview, "Each BlueDBM node consists of flash storage coupled with a field-programmable gate array (FPGA), and is plugged into a host system's PCIe port. Each node is connected to up to eight other BlueDBM nodes over a high-speed serial link capable of 10-gigabit bandwidth at 0.5ps latency. By default, the FPGA includes platform functions such as flash, network and on-board DRAM management and exposes a high-level abstraction."

Photonic memory. In a paper published in *nature.com* last year, a group of scientists from English and German universities proposed an integrated all-photonic, non-volatile multi-level memory. Photonic data storage can greatly improve the performance of computing systems by reducing the latencies associated with electrical memories and potentially eliminating optoelectronic conversions. While scientists have experimented with photonic memory earlier, these attempts mostly resulted in volatile memory.

The recent breakthrough uses phase-change materials to achieve a robust, non-volatile, all-photonic memory. In simple terms, this solution uses waveguide technology to move light from lasers across a germanium, tellurium and antimony alloy nano-coating. The structure of the alloy is altered in predictable patterns when hit by light from a high-intensity laser. A low-power laser is used to read the patterns and translate it as data. By using optical near-field effects, the team has achieved bit storage of up to eight levels in a single device that readily switches between intermediate states. According to the paper, the on-chip memory cells feature

single-shot readout and switching energies as low as 13.4pJ at speeds approaching 1GHz.

Before they think of commercialisation, the team has to figure out how to reduce the size of their prototype, which is much larger than current-generation memory chips! However, in terms of performance, photonic memory can easily outperform SSDs, so it is a worthy pursuit.

SSDs cannot be monarchs

Speaking on the future of SSDs at last year's SNIA Storage Developer Conference in Santa Clara, California, Jim Handy from Objective Analysis said, "Future memory and storage systems will include everything: DRAM, non-volatile memory, NAND and hybrid hard drives. One technology will not kill off the others."

Jim added, "The result of all the changes in technologies and architectures is a new component: storage-class memory. This memory will combine the benefits of a solid-state memory with DRAM performance and robustness and archival capabilities and low cost of hard-disc magnetic storage. The persistent memories will disrupt the entrenched thoughts on possible and available latency budgets for storage... The computer of tomorrow will have fixed DRAM size made of stacked packaging of DRAM dice, and upgradeable non-volatile memory that will be the equivalent of DIMMs. The storage system will include both flash and disk with the flash on PCIe of its own bus. The magnetic drives will continue to exist for mass storage as there is no foreseeable price crossover for high-density, long-term storage. Finally, the storage class memory software will eventually get to the point where it will contribute to overall system performance."

So it is clear that, while we will see more SSDs around us in the coming years, an all-flash era might be quite far away. **EFY**



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